However, this does not mean that you know the information in the *simplest* way. In particular, you understand the frequency response as a set of values that change with frequency. Just as in our train analogy, the frequency response can be more easily understood in terms of the terrain *surrounding* the frequency response. That is, by the characteristics of the s-plane.

With the train analogy in mind, look back at Fig. 32-3, and ask: how does the shape of this s-domain aid in understanding the frequency response? The answer is, it doesn't! The s-plane in this example makes a nice graph, but it provides no insight into why the frequency domain behaves as it does. This is because the Laplace transform is designed to analyze a specific class of time domain signals: *impulse responses that consist of sinusoids and exponentials*. If the Laplace transform is taken of some other waveform (such as the rectangular pulse in Fig. 32-3), the resulting s-domain is meaningless.

As mentioned in the introduction, systems that belong to this class are extremely common in science and engineering. This is because sinusoids and exponentials are solutions to *differential equations*, the mathematics that controls much of our physical world. For example, all of the following systems are governed by differential equations: electric circuits, wave propagation, linear and rotational motion, electric and magnetic fields, heat flow, etc.

Imagine we are trying to understand some linear system that is controlled by differential equations, such as an electric circuit. Solving the differential equations provides a mathematical way to find the impulse response. Alternatively, we could measure the impulse response using suitable pulse generators, oscilloscopes, data recorders, etc. Before we inspect the newly found impulse response, we ask ourselves what we *expect* to find. There are several characteristics of the waveform that we know without even looking. First, the impulse response must be *causal*. In other words, the impulse response must have a value of zero until the input becomes nonzero at $t = 0$. This is the cause and effect that our universe is based upon.

The second thing we know about the impulse response is that it will be composed of *sinusoids and exponentials*, because these are the solutions to the differential equations that govern the system. Try as we might, we will never find this type of system having an impulse response that is, for example, a square pulse or triangular waveform. Third, the impulse response will be *infinite* in length. That is, it has nonzero values that extend from $t = 0$ to $t = +\infty$. This is because sine and cosine waves have a constant amplitude, and exponentials decay toward zero without ever actually reaching it. If the system we are investigating is **stable**, the amplitude of the impulse response will become smaller as time increases, reaching a value of zero at $t = +\infty$. There is also the possibility that the system is **unstable**, for example, an amplifier that spontaneously oscillates due to an excessive amount of feedback. In this case, the impulse response will *increase* in amplitude as time increases, becoming infinitely large. Even the smallest disturbance to this system will produce an unbounded output.